CHAPTER A

Reasoning and preparing the test

A.1 Logic: From words to evidence

Sets and proposals

$x \in A$ means that x is an element of A; $A \subseteq B$ means that each element of A belongs to to B. Empty assembly is noted $\varnothing$. The intersection corresponds to "and", the meeting to "or" inclusive. In an E universe, $E \setminus A$ is the complement of A. A couple ($a; b$) is ordered ; $A \times B$ group the couples whose first component is in A and the second in B. The Cardinal a finite set is its number of elements.

Example solved. For $A = \{1; 2\}$ and $B = \{2; 3\}$, intersection is $\{2\}$, the meeting $\{1; 2; 3\}$. The product $A \times B$ contains (1; 2) , $(1; 3), (2; 2),$ (2; 3): Cardinal 4.

Numbers and intervals. $\mathbb{N}$ contains here 0 and natural integers; $\mathbb{Z}$ the relative integers; $\mathbb{D}$ the Decimals with finished writing; $\mathbb{Q}$ quotients of two integers, with a non-zero denominator; $\mathbb{R}$ the real. We have $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{D} \subset \mathbb{Q} \subset \mathbb{R}$. Thus $\frac{1}{3}$ is rational but not decimal; $\sqrt{2}$ is real not rational (previous result). [-2; 3[means $-2 \leq x < 3$; an infinite bound is never included.

An implication "if P, then Q" expresses that P is a sufficient condition for Q and Q a necessary condition for P. His converse is "if Q, then P". His contraposition is "if no Q, then not P'; it has the same logical value as involvement. An equivalence brings together the two implications.

Example solved. "If x = 2, then x^2 = 4" is true. Converse is false for $x = -2$. Equality x = 2 is sufficient for x^2 = 4, but not necessary. Contrapositive "if $x^2 \neq 4$, then $x \neq 2$" is true.

To refute a universal proposal, a counter-example is enough. To prove that an object exists, one can be built and its properties checked. Nier "for all x, P(x)" gives "it exists x As not P(x)"; denying "there is x, P(x)" gives "for all x, not P(x). quantifiers are not payable; the meaning of the sentences is.

Example solved. The denial of "for all real x, $x^2 > 0$" is "there is a real x such that $x^2 \leq 0$". The choice x=0 checks this denial.

Disjunction of cases. Solve $|x|$ = 2 by distinguishing x positive or zero and x negative: 2 and -2. Cases cover the whole area.

I don't know. Suppose that x^2 + 1 = 0 has a real solution. $x^2 = -1$, in contradiction with the positivity of a square. So there is no real solution.

Identity, equation, parameter. $(x + 1)^2 = x^2 + 2x$ + 1 is a true identity for all real x. x^2 = 1 is an equation of which x is unknown. $x^2 - 2mx$ + 1 = 0, m is a fixed parameter when looking for x; discussing according to m is another level of reasoning.

Cross-cutting exercises

1. Deny: "There is a real x such as $x^2 < 0$". Is the original proposal true?

2. Give the converse and the contrast of: "If an integer is multiple of 4, then it is even". Which ones are true?

3. For $A = \{0; 1\}$ and $B = \{1; 2; 3\}$, calculate $A \cap B$, $A \cup B$, $A \times B$ and their cardinals.

4. To show by the absurd that there is no larger natural integer.

5. Sort -2, $0{,}125$, $\frac{1}{3}$, $\sqrt{2}$ in the smallest usual set; translate $x > 1$ and $x \leq 4$ by one interval. Specify the status of x and m in $mx + 1 = 0$. Solve according to m.

Cross-sectional corrections

1. Negation: for all real x, $x^2 \geq 0$, true; the original proposal is false.

2. False converse, counter-example 2. Contrapositive: if an integer is not even, it is not multiple of 4; true.

3. Intersection $\{1\}$, Cardinal 1 ; meeting $\{0; 1; 2; 3\}$, Cardinal 4 ; produced by six couples $(0; 1), (0; 2), (0; 3), (1; 1), (1; 2), (1; 3)$.

4. If N were the largest, N+1 would be a strictly larger natural integer: contradiction.

5. Respectfully $\mathbb{Z}$, $\mathbb{D}$, $\mathbb{Q}$, $\mathbb{R}$; interval]1; 4]. x unknown, m parameter. If $m \neq 0$, only solution $-\frac{1}{m}$; if m=0, the equation becomes 1 = 0, no solution.

A.2 Python: understanding what is being done

The files provided run with Python 3. A function receives arguments and returns a result; a loop repeats an action. A program variable can change value by assignment, unlike a fixed unknown in an equation.

Lists and indices

`L=[1,4,9]` defines an extended list. `L.append(16)` add an element; `del L[0]` delete the first. `L[0]` reads the index element 0 and `len(L)` gives the length. A list in understanding, such that `[k*k for k in range(5)]`, contains squares from 0 to 4. are part of the list: it is not a set.

Example solved. After `L=[1,4,9]`, `L.append(16)` then `del L[0]`, the list is [4.9,16]. `[x for x in L if x%2==0]` [4,16]. The test is a logical proposal executed for each element.

File	Uses and controls to be performed
`suites.py`	Terms, amounts, thresholds, factorial, Fibonacci, Syracuse, manipulations test n=0, q=0, q=1, threshold already reached and threshold inaccessible.
`analyse.py`	Drying rate, Newton, Euler, exponential threshold. Avoid not zero and derivative zero; distinguish stop and convergence.
`probabilites.py`	Law, moments, Monte-Carlo, samples, frequency of letters. sum of probabilities, non-zero size and simulation seed.

`archimede.py`	Polygons inscribed and circumscribed, without using pi in construction. Control the tightening of the bounds.
`verifier.py`	Execute deterministic controls and limit cases. It is not a demonstration of a theoretical value.

Practical work to be distributed

1. Sequences: generate ten terms, compare loop and formula, then calculate cumulation. length of the list.
2. Cuts: take three positive steps and three negatives; compare to algebraic rate calculation.
3. Monte Carlo: compare 1,000 and then 10,000 points, with two seeds; comment on the fluctuation.
4. Samples: simulate N=1,000 size samples n=10 then n=100; compare mean Observed and expected. Count averages in threshold $\frac{2\sigma}{\sqrt{n}}$.
5. Euler and Archimedes: produce a table of results with several resolutions; indicate what It is observed and demonstrated.

These TPs are alternatives distributed over algorithmic and consolidation sessions. The results of a simulation depend on the seed; the control report provides an example of reproducible execution.

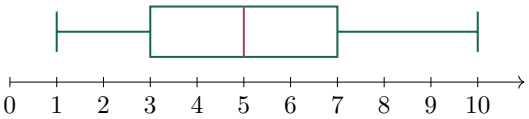
A.3 Read and represent data

A weighted average is $\frac{\sum n_i x_i}{\sum n_i}$. The median shares an ordered series; for a workforce The average of the two central values is used here. $\frac{n}{4}$ and $\frac{3n}{4}$ are explicitly used when values are requested. The descriptive variance uses the denominator n, without correction of estimate.

In a bar diagram, the height represents the number or frequency. In a histogram With unequal width classes, it must be proportional to the area of the rectangle. circular distributes the 360 ° A mustache box represents indicators of position and dispersion, not the frequency of each value. Always read origin, units and scale before you compare.

Example solved. Two classes of width 2 and 4 each contain 20 observations. 10 and 5 give in both cases an area of 20. Equal heights would make the second class artificially twice as important.

Immediate application. Categories A, B, C are effective 10, 20, 30. Build a diagram in bars and calculate the angles of a circular diagram. For series 1,1,3,7, give average, Median, extended and variance. Read the box below: minimum, quartile, median and maximum. Can we find all the data?



Independent box from the previous series; moustaches at extreme values.

Corrected. Box: minimum 1, first quartile 3, median 5, third quartile 7, maximum 10. These five indicators do not determine all observations. °, 120°, 180°. Bar heights are 10.20.30. Series: mean 3; median 3; extent 6; variance $\frac{4+4+0+0+16}{5} = \frac{24}{5}$.

A.4 Annales: read the format without reducing the program

The topics of 2026 were consulted to identify the steps requested.
the first edition of the EAM, before the application of the 2026 program; they serve as training after
The original subjects EV do not present themselves as subjects
official.

Source	Observed approaches	Reinvestment
Metropolis 2026, 26MATSPEGEME 1	Tree with unknown probability; assertions justified; geometry linking angle and equation.	C05, C03, C09; EV2-A, EV3A and EV4-B.
Official subject 2026, 26MATSPEGEG 11	Discreet model, cumulation and reading of loop; coordinates and angle; tangent and quotient.	C01, C03, C06; EV1-T1, EV2-T1.
Caribbean Guiana 2026, 26-MATSPEGEAG1	Threshold of a sequence; tree and random rate ; exponential and true/false product.	C01, C05, C08, C09; EV4-T1 and EV4-B.
Zero subjects Education	Fluency, independent problems and writing without calculator.	Training T1/T2 of each lot.

The exact links are provided in the file `SOURCES.md`. Subjects are not reproduced in full.
To work an annual, note the code of the subject, exercise and questions actually
study; correct reasoning, not just results.

Prepare two hours without calculator

Start by reading the two parts; set aside about 20 to 25 minutes for fluency.
exercise, write the assumptions and domain before calculation. If a step blocks, use the result
to keep a dozen minutes to check the signs, the signs,
the units and answers to the questions. These benchmarks should be adapted to the rhythm
during training.